

Offline Diagnostics for Plugin-Based Agent Harnesses: Design, Check Lifecycle, and Field Experience with DeepSeek Harness

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Abstract

DeepSeek Harness (dsh) is a plugin-based agent harness whose central design tenet — “*everything is a plugin*” — is realized on Cordis, a meta-framework of *spatiotemporal composability* [2], [3]. While the architecture makes every capability (model adapters, tools, sessions, the agent loop itself) replaceable at configuration time, it also inherits a systemic fragility: a single malformed patch, a duplicate entry id, a shadowed module instance, or a corrupted session log can brick the profile at boot or stall the entire web server with little or no diagnostics. This paper reports on the design, implementation, and field experience of *dsh-doctor*, an offline diagnostic for this failure space, and derives from it a *check-lifecycle model*: (i) checks as declarative data distributed through a remote catalog; (ii) introspection of installed harness contracts instead of hard-coded assumptions; (iii) a fixture-based certification gate that proves a check does not false-positive and does catch its target; and (iv) an evolution loop from community report to certified check. We document 34 checks mapped to 33 community failure reports, 109 regression tests that have caught real bugs in the tool’s own history, a three-round false-positive elimination campaign, and the convergence of three independent community tools on a shared machine-readable contract. We conclude with open problems for the broader ecosystem: check registry governance, certification matrices across release trains, and the role of formal composability guarantees in preventing whole failure classes.

Keywords: agent harness, plugin systems, offline diagnostics, composability, check lifecycle, DeepSeek Harness, Cordis

1. Introduction

Modern AI agent harnesses have moved from monolithic loops to composable plugin architectures. DeepSeek Harness (dsh) operationalizes “*everything is a plugin*”: model adapters, tool registries, session storage, sandboxing, the agent loop, and even the web UI are all plugins composed at runtime by Cordis, a plugin framework formalized in an 88-page paper on *spatiotemporal composability* [2]. The dsh repository has attracted over 110,000 stars and 10,000 forks within days of its public release [1], and a community ecosystem of plugins and marketplaces is forming around it at high velocity.

This architectural bet carries a distinctive failure mode. In a harness where configuration replaces code as the composition mechanism, the *deployment-time* state — profile manifests, bundle patches, the composed plugin tree, the append-only session logs — becomes the primary locus of fragility. Empirically, this is not hypothetical: the community has reported a recurring family of failures in which a single bad install bricks the profile at boot, stalls tool execution, or makes sessions unrecoverable, often with opaque errors that point at nothing actionable [7], [8], [11], [13], [17], [18], [19]. The harness’s own `--dump-config` never mounts the loader, so it passes on broken setups; the failure class was consolidated into an advisory, “*the plugin-install path needs guardrails*” [6].

This paper makes the following contributions:

1. **A taxonomy and empirical corpus.** We consolidate 18 community failure reports into a three-group taxonomy (environment, profile/composition, session-log integrity), each entry verified against the vendored harness source and, where possible, reproduced with synthetic fixtures.

2. **The design of dsh-doctor.** A zero-dependency, single-file offline diagnostic that runs before boot and reports which of the known failure classes will bite, as a CLI and as a dsh plugin with a settings panel and a read-only JSON API.
3. **A remote check catalog (checks-as-data).** New checks are declarative rules shipped as data and distributed without a plugin release, with a security model that guarantees the remote payload can never execute code.
4. **A check-lifecycle model.** We argue that checks themselves rot as the harness evolves, and propose four properties — checks-as-data, introspection, a fixture-based certification gate, and an evolution loop — each backed by a running reference implementation.
5. **Field experience.** We document 34 checks, 109 regression tests, real bugs the corpus caught in the tool’s own history, a three-round false-positive elimination campaign, and the convergence of three independent community tools on a shared **dsh-doctor/v1** output contract.

The remainder of this paper is organized as follows. Section 2 provides background on the harness architecture and the threat model. Section 3 surveys related work. Section 4 details the system design. Section 5 presents the check-lifecycle model. Section 6 evaluates the approach empirically. Section 7 discusses limitations and open problems. Section 8 concludes.

2. Background and Threat Model

2.1 DeepSeek Harness and Cordis

DeepSeek Harness is an open-source (MIT) agent harness whose products — web app, headless runner — are composed entirely from plugins [1], [4]. The composition framework is Cordis [3], which the harness *source-vendors* into its own repository under the scope `@deepseek-ai/cordis` (version 4.0.1, based on upstream 4.0.0-rc.7, with local hardening patches) [4], [24]. Cordis’s model fits in five ideas: plugins as objects contributing capabilities; a context as a store of named capabilities; declared rather than ordered dependencies; typed events with declared dispatch; and teardown — every registration knows how to undo itself [24], [25]. The formal foundation [2] lifts classical effect/coeffect concepts to runtime mechanisms: *reversible effects* (every context transformation carries a tracked inverse) and *reactive coeffects* (context changes notify components), unified into a single context type; the calculus establishes spatiotemporal composability from a single component to interleaved systems.

In implementation, a *Fiber* is the per-instance lifecycle state machine of a plugin (PENDING → LOADING → ACTIVE → FAILED → DISPOSED → UNLOADING); a plugin that declares `inject` dependencies waits in PENDING until they are resolvable [2], [24]. `ctx.effect(fn)` registers a side effect with a disposer; `ctx.provide` and `ctx.on` build on it. Two properties matter for our threat model:

- **Declared dependencies are not enforced at the OS level.** `inject` constrains what a plugin can *see* through the context, not what the process can do; a plugin that fails to declare a dependency may nonetheless activate before the service is ready (or never, if the service never appears in its scope), producing either a permanent PENDING or an undefined-service crash [17], [19].
- **Composition is configuration.** The composed tree is decided by profiles, bundles, and patches — text files that any install path can corrupt or duplicate [4], [13], [19].

2.2 Profiles, Bundles, and Patches

The harness defines three configuration layers [4]:

- **Bundle** — distributes a set of Cordis configuration and the plugin code that mounts them, declared via `dsh.bundle.patch` in the package manifest.
- **Profile** — a named composition listing which bundles to stack (web and headless ship as templates), stored under the harness home (`~/.dsh/profiles/<name>`).

- **Patch** — replaces or inserts configuration rows, applied in order: bundles (in declaration order) → profile `cordis.patch.yml` → home-level patch → CLI `--patch` overlay. A patch locates an entry by id and replaces its whole config, or inserts a new row.

Because “the last layer wins” and rows are keyed by id, **duplicate ids across layers are a hard boot failure**: the loader’s `EntryGroup.update` throws `duplicate loader entry id` [7], [19]. This is the first and most frequently reported crash class.

2.3 Append-Only Session Logs

A session is an append-only event log (typically `session.jsonl.zstd`) with monotonically increasing `seq` values; the model reads a *projection* of the log, and the invariant “*model-visible means logged*” guarantees everything the model sees can be reconstructed from the log [4], [24], [25]. The log is also the source for resume, fork, audit, and token metering. Threats to log integrity include: `seq == index` gaps and duplicates [9]; post-**end-seed** replay of a committed tail [12]; unknown event types without **ignorable** causing wholesale refusal [14]; single-frame zstd containers that make `session.list` fail wholesale [16]; and `sourceEventSeqs` referencing non-earlier events after compaction [15].

2.4 Threat Model

We define the diagnostic target as *deployment-time state*: environment (PATH, node version, native binaries, storage, ports), profile composition (manifests, patches, resolved bundles), and session logs (structure, continuity, integrity). The diagnostic must run **offline** (before boot, when the harness itself may be unusable), be **zero-dependency** (no harness internals required), and produce **actionable** output (what to fix, not just that something is wrong). Attacks or accidents are out of scope for prevention; the tool’s contract is detection and remediation guidance.

3. Related Work

3.1 Plugin-System Fragility

Plugin systems have long traded composability for deployment fragility. VS Code’s extension host, for example, defers full uninstall cleanup of executable extensions to process restart, a pattern the Cordis paper explicitly cites as motivation for component-level revertible effects [2]. Koishi [5], the chatbot framework that predates and uses Cordis, demonstrates the same composition model at scale but in a community where plugin authors control their own installs. The distinctive property of the dsh ecosystem is that *the harness itself is assembled from plugins*, so composition failures are boot failures, not feature degradations.

3.2 Offline Diagnostics for Developer Tooling

Offline/static diagnostics are well established in compiler and package-manager tooling (e.g., `npm doctor`, `cargo doctor`, IDE linters). What is novel here is the *object* of diagnosis: not source code or dependency graphs, but a composed runtime tree whose contracts (event-type tables, module symbols, patch grammar) live in an installed, versioned artifact. The closest analog is runtime anchor verification, which we develop further in Section 5.

3.3 The Emerging dsh Diagnostic Ecosystem

Within weeks of the harness release, several independent diagnostic tools appeared: `dsh-plugin-doctor` (author-side pre-publish checks, `profile-shadow` tripwire) [21]; `dsh-diagnose` (symptom-based diagnosis with release-anchored knowledge) [22]; cicero yang’s zero-dependency CLI [30]; boyin111-1’s sibling offline diagnostic [31]; plus session-focused tools (`dsh-session-doctor`, `dsh-session-repair`). A proposal for an official **dsh doctor** command and a community registry contract has been consolidated in [20], and a broader RFC covering registry v2, `dsh plugin check`, and `dsh doctor` in [32]. This paper’s contribution is distinct: the

check-lifecycle layer underneath all of these — how any check, in any of these tools, is born, proven correct, distributed, and kept honest across harness upgrades [33].

3.4 Formal Foundations

The Cordis paper [2] provides the theoretical backing for the harness’s composition semantics and, we argue in Section 5, for the *diagnostic* layer as well: the same failure to achieve revertible effects and reactive coeffects that the paper formalizes is what manifests in the field as leaked registrations [10], [11] and silent contract drift [8].

4. System Design: dsh-doctor

4.1 Architecture

dsh-doctor is a single-file, zero-dependency Node.js program (currently ~1,050 lines) that also ships as a proper dsh plugin: the plugin’s server route shells out to the same CLI with `--json`, so the CLI, the settings panel, and the HTTP API (`GET /dsh-doctor/run`) share one source of truth [26]. The repo-root entry is a thin wrapper for `node dsh-doctor.mjs` compatibility.

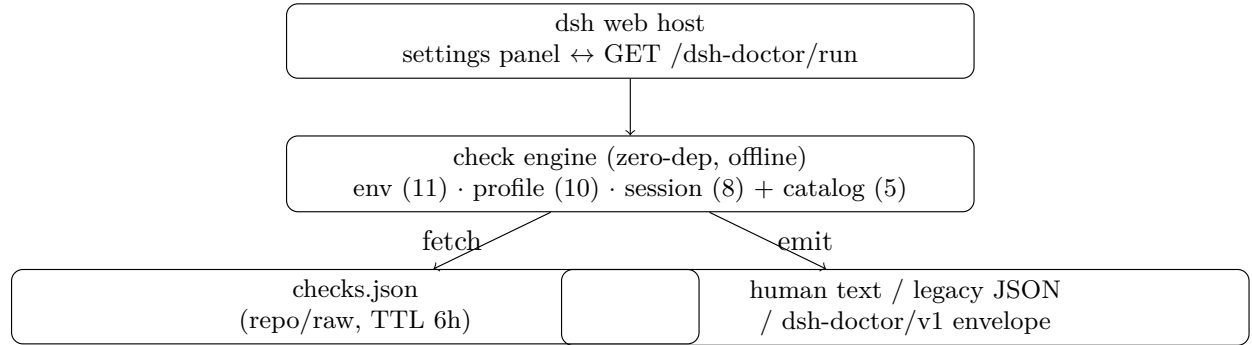


Figure 1: dsh-doctor architecture: one check engine, three output modes, remote catalog with TTL cache.

The tool runs three groups of checks — **env**, **profile**, **session** — plus a remote catalog layer, and emits either human-readable output or machine-readable JSON. Exit codes: 0 all pass, 1 problems found (or any WARN in envelope mode), 2 any FAIL in envelope mode.

4.2 The 30 Checks

As of this writing the tool ships 25 built-in checks and 5 catalog checks (Section 4.3), each mapped to one or more community failure reports:

Environment (7 built-in + 4 catalog): E1 binaries on PATH (`node/pnpm/zstd`) [#1270]; E2 `.env` is a file not a directory [#71]; E3 node version and `--expose-internals` reachability [#113], [#1313]; E4 node-pty native binary presence [#1219]; E5 storage JSON validity (strict UTF-8 + parse) [#1357]; E6 **anchor tripwire** — verifies the contracts our S6/S7/S10 checks rely on still exist in the installed `dsh-session` [9]; E10 web port 3080 availability (dsh web itself = normal, foreign process = FAIL) [#1719] [20]; E7 `dsh` on PATH (catalog); E8 profile `.npmrc` workspace flag (catalog, warn) [23]; E9 `config/workspace.json` validity (catalog) [#1357]; E11 `settings.yaml` writability / sudo ownership (catalog) [34].

Profile (10 built-in): P1 bundle resolution [#917]/[#1377]/[#880]; P2 bundle-vs-user-patch insert id collisions (the `duplicate loader entry id` crash) [#1404] [7]; P3 user-patch insert names resolvable from the profile anchor [#1197]/[#880]; P4 `file`: dependencies intact [#1197]; P5 top-level `@deepseek-ai/*` duplication (dual module instances) [#1486] [8]; P7 `cordis.patch.yml` structural lint (~ `insert`: null literal, tab indentation, missing colon, top-level mapping+sequence mixing) [#1724] [13]; P8 adapter-provider

registration conflicts (`DUPLICATE_ADAPTER`) [#1904] [35]; P9 `ctx.settings` usage without declared settings dependency [#1904] [35]; P10 inject referencing client-only services (`@deepseek-ai/dsh-client-*`) → permanent PENDING [#1947] [17]; P11 installed bundle’s `main` entry artifact missing (unbuilt source tree) [#1965] [18].

Session (8 built-in): S1 orphan `tool_call` [#1363]/[#1544]; S2 unclosed turns [#466]/[#1265]; S6 `seq == index` continuity with chunk expansion [#1333]/[#1452]/[#1469]; S7 `post-end-seed` replay [#1497]; S8 unknown event types without `ignorable` [#1538]; S9 `zstd` container frame count [#1043]; S10 `sourceEventSeqs` drift [#1469]; S11 whole-session scan: corrupt → quarantine, oversized → cold-start heap risk [#1550].

Each built-in check is validated by at least one synthetic fixture asserting isolation: on a *bad* fixture the target check must fail while every other check passes, and healthy baselines must not false-positive. This discipline is what makes the corpus useful for the certification model of Section 5.

Table 1 — Check inventory (34 checks, 33 mapped community reports, as of v0.4.5).

Table 1: Check inventory (34 checks, 33 mapped community reports, v0.4.5).

Group	IDs	Count	Coverage class	Representative reports
env (built-in)	E1-node/pnpm/zstd, E1-E6, E10	8	PATH / files / native binary / storage / anchors / port	[#1270] [#71] [#113] [#1219] [#1357] [#1534] [#1719]
env (catalog)	E7-E9, E11	4	declarative probes (path/JSON/text/writable)	[#1270] [23] [#1357] [34]
profile	P2-P5, P7-P11, P13, P14, P15, installed_bundle	13	composition: resolution / collisions / patch syntax / dual-instance / inject semantics / artifacts	[#1404] [#1486] [#1697] [#1724] [#1904] [#1947] [#1965]
session	S1, S2, S6-S11	8	log integrity: continuity / replay / types / containers / drift / heap	[#1363] [#466] [#1333] [#1497] [#1538] [#1043] [#1409] [#1550]

Coverage honesty: two symptom families (sandbox denials, approval policy) have **no** offline probe and are explicitly documented as gaps in the published symptom→check mapping [26], [34].

4.3 Layer A: Remote Check Catalog (Checks-as-Data)

The catalog (`plugin/checks.json`) holds declarative rules — data, not code. The engine executes only read-only probe primitives (`command-exists`, `path-exists/is-dir/is-file`, `json-valid`, `text-contains/text-not-contains`, `file-size-above`, `glob-count`, `file-writable`), so a remote payload can never execute code; it can only add pattern checks. Each run fetches the catalog from the repository (3 s timeout), caches it under the harness home (TTL 6 h), and falls back to the last-known-good cache, then the bundled copy. New checks therefore reach every installed instance within 6 h of being committed upstream, with **zero plugin releases** — five catalog checks have shipped this way to date. This property (Section 5, property 1) is the distribution mechanism that makes the ecosystem’s check inventory updateable without the “update the tool” tax.

4.4 Layer B: Self-Update

The tool also watches its own npm version (`dist-tags.latest` vs. installed), with the same TTL/fallback discipline. It never touches the installation unasked: `--update` runs `pnpm install` in the hosting profile, `DSH_DOCTOR_AUTO_UPDATE=1` enables opt-in automatic updates, and an honest boundary is documented — Cordis loads plugins at boot, so the new engine activates only after a restart.

4.5 Output Contract: `dsh-doctor/v1` Envelope

To make diagnostics consumable by other tools (CI, marketplaces, symptom tools), `dsh-doctor` emits — alongside its legacy JSON — a v1 envelope [27]:

```
{
  "schema": "dsh-doctor/v1",
  "tool": "dsh-doctor",
  "generatedAt": "ISO-8601",
  "profile": "<name or dir>",
```

```

    "exitCode": 0,
    "summary": { "pass": 2, "warn": 0, "fail": 0 },
    "ok": true,
    "checks": [ { "name": "<check id>", "status": "pass", "detail": "..." } ]
}

```

with lowercase `pass|warn|fail` status vocabulary, exit codes 0/1/2 (all-pass / any-WARN / any-FAIL), and a documented minimal compatible subset `{ok, checks:[{name,status,detail}]}` that any consumer of the full envelope can read [27]. Two other community tools have adopted this contract (Section 6.4).

5. The Check-Lifecycle Model

The RFC consolidating the ecosystem’s tooling [32] standardizes the *tools*: registry contract, publish gate, doctor command. What remains unspecified is the **lifecycle of the checks themselves**. We argue this is a first-class problem, because checks rot — and false-positive.

Checks rot. Every check encodes assumptions about harness internals: event-type tables, seq semantics, patch grammar, file layout, module symbols. Every harness release can silently invalidate them. Field evidence: the anchor-tripwire check E6 exists *precisely* because of this; the dual-module-instance crashes [#1486]/[#1697] [8], [11] and the process-singleton collision [#1415] [10] were contract-drift failures no static list caught early; and our own P3 shipped silently dead for months — a wrong regex matched nothing, so it always “passed” — until fixture testing exposed it (Section 6.1).

Checks false-positive. Static checks without semantic boundaries misfire: a substring match on `npm run` misfires on `pnpm run` [13]; a name match on `exec` misfires on `node:sqlite’s DatabaseSync.exec` [36]; three rounds of P9 false positives were eliminated in this work (Section 6.2). A noisy check poisons the credibility of the entire tool — worse than no check.

We therefore propose four properties, each backed by a running reference implementation in dsh-doctor:

Property 1 — Checks are data, not code. A check is a declarative, read-only probe definition plus its target contract anchor, its good/bad fixture pair, and the release train it was certified on. Distribution happens through a *check registry* (not a plugin registry) with TTL cache and offline fallback. *Reference*: the Layer-A catalog (Section 4.3), which has shipped 5 checks with zero releases.

Property 2 — Introspection instead of hardcoding. Read the installed harness’s contracts (event-type table, module symbols, patch grammar) to derive or validate expectations. On upgrade, either the anchor passes (check still valid) or the tripwire fails loudly (no silent rot). *Reference*: S8 parses the installed `KNOWN_SESSION_EVENT_TYPES` with fallback; E6 verifies the chunk-expansion contract (the `expandRow/seq0+k` semantics S6 depends on), the `session/end-seed` literal, and the `sourceEventSeqs` field still exist — *wherever* the current package keeps them (in `dsh-session` for the rc-line, in the `dsh-session-format-v0-to-v1` migration package after the 0.1.5 split). The same idea appears independently in the knowledge domain as dsh-diagnose’s `check-knowledge-anchors` [22].

Property 3 — Certification gate. A check enters the ecosystem only if its fixture pair proves *good-does-not-false-positive*, *bad-gets-caught* on the declared train. This upgrades shape-level contract checks to *correctness-level* certification. *Reference*: the 109-test corpus whose isolation property has caught real bugs (Section 6.1), and the fixture-based acceptance harness proposed in [32].

Property 4 — Evolution loop. Community report → candidate check → fixture certification → catalog distribution → auto-update. The diagnostic layer itself “evolves on the fly” — the paper’s theme [2] applied to the diagnostics. *Reference*: the report-driven addition of P8–P11 (Section 6.1) plus the Layer-B update mechanism.

We also propose a check-entry schema:

```

{
  "id": "E11-settings-writable",

```

```

"section": "env",
"severity": "error",
"probe": { "type": "file-writable", "path": "{home}/settings.yaml", "required": false },
"anchor": { "package": "@deepseek-ai/dsh-settings", "symbol": null, "train": "0.1.0-rc.6" },
"fixtures": { "good": "path-or-inline", "bad": "path-or-inline" },
"certifiedOn": "0.1.0-rc.6"
}

```

A working draft of this model is published as [33].

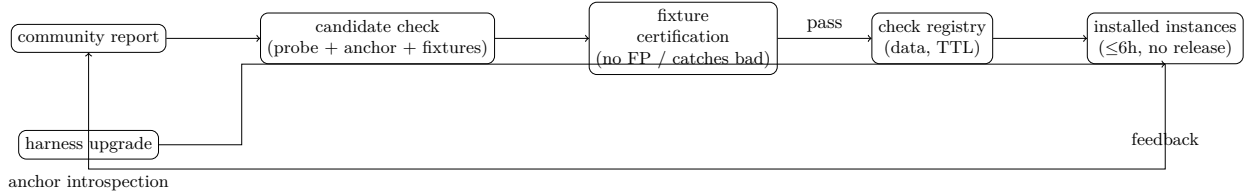


Figure 2: The check lifecycle: report → candidate → certify → distribute → verify (property 4), with introspection (property 2) guarding every upgrade.

6. Empirical Evaluation

6.1 Detection: Real Bugs Caught by the Corpus

The fixture corpus has repeatedly caught real defects:

- **P3 silent no-op.** The original regex for user-patch insert names matched a patch format that does not exist in the wild (the real format indents `name:` under `- id:` without a dash). The check “passed” for months without ever checking anything; a fixture built from the real format exposed it.
- **P6 regex overmatch.** The `text-not-contains` probe’s pattern let `\s` consume the separator space, so any `name:` line matched — flagging every plugin. Semantic correction (`[^"\s]+` first, then an internal space) fixed it.
- **P8–P11 genesis.** All four profile checks were added in response to *reported* failures: adapter-provider conflicts [#1904], missing `settings` inject [#1904], client-only service injects [#1947], and unbuilt `main` artifacts [#1965]. Each addition was itself validated by new fixtures (now 64 total).
- **Field validation.** A user’s independent fix of a duplicate-id boot crash [19] turned out to match P2’s exact detection shape, and a genuine `ctx.settings-without-inject` defect was confirmed in the same thread — validating both P2 and P9 against real incidents.

6.2 The Anchor Tripwire in Action: A Cross-Train Format Migration

Twenty-seven days after the first field deployment, the harness had moved from `0.1.0-rc.6` to `0.1.5-rc.1` and, in doing so, changed two things at once: it migrated from an `npx-checkout` install to a **global npm install**, and it split the session format into a versioned chain (`v0→v1→v2→v3`) whose migration logic lives in separate packages (`dsh-session-format-v0-to-v1`, `-v1-to-v2`, `-v2-to-v3`). Running the unchanged tool against the new install reproduced, in one afternoon, exactly the failure mode Property 2 exists to prevent:

- **The locator went blind.** E6 and S8 resolved `dsh-session` only through the `node_modules/.bin/dsh` (`npx`) layout. Under the global install the lookup failed, so both checks fell back to a built-in event-type table captured at rc.6 — E6 degraded to a green “anchors unverified” state, and S8 compared sessions against a stale vocabulary.
- **The consequence was 49 false positives.** Against the stale table, S11 reported 49 sessions as corrupt because their legacy event types (`assistant/chunk`, `tool/code-dispatch`) are absent from the

current `KNOWN_SESSION_EVENT_TYPES` — yet the harness reads them correctly through the migration chain. A check that had been right on its certified train became a false-positive generator on the next one.

- **The anchor moved rather than disappeared.** The `expandRow/seq0+k` expansion contract S6 depends on is no longer in `dsh-session`; the equivalent disposition table now lives in `dsh-session-format-v0-to-v1`. E6 *failed loudly* on the missing anchor instead of reporting green — and that failure is what directed the repair.

The fix was structural rather than a one-off patch: `dsh-session` resolution now covers `npx`, `global`, and `profile` layouts; when several installs coexist, the newest supplies the “current” contract while the **readable** event-type set is the union across installs plus the types recognized by the installed migration packages (54 current + 15 legacy = 69 on the observed machine, versus 41 in the stale table); E6’s anchors were re-pointed to wherever the current release keeps them; and a failed location is now reported as `skip` (“does not apply”) rather than a silent pass. Regression fixtures for all three behaviours were added with the fix (109 tests total, all passing).

Two lessons generalize beyond this ecosystem. First, **introspection must itself be version-robust**: reading the installed contract beats hardcoding only if the contract can still be *found* after the install layout changes. Second, **the tripwire’s value is in failing loudly**: the brief window in which E6 reported “unverified” as a pass is precisely the window in which the tool would have shipped confident wrong answers. The same check infrastructure now also carries a 26-check security layer (`dsh-security`), evidence that the lifecycle model is not specific to diagnostics.

6.3 False-Positive Elimination: Three Rounds on One Check

P9 (missing `settings` dependency) was flagged on the author’s own production profile against three plugins. Manual source verification showed all three were false positives, each a distinct class:

1. **Substring collision.** `sctx.settings` (a different variable) contains `ctx.settings` as a substring — the same trap as the `npm run pnpm run` substring miscount we made while verifying the harness repo’s own `package.json` (discussion #1702, a correction we later acknowledged publicly).
2. **Wrong artifact scope.** A client-side bundle (`client.js`) was being scanned as server code.
3. **Wrong declaration.** `String.match` returned the *first* `inject = [...]` array in a bundled file — an internal module’s declaration — masking the plugin’s own `inject` (which correctly included `settings`).

Each was fixed with a targeted mechanism (negative lookbehind; excluding `client/web` directories; collecting all `inject` declarations and satisfying on any). The same lesson — *name-based static checks need semantic boundaries* — is echoed by the ecosystem’s `DatabaseSync.exec` false-positive report [36].

6.4 Real-Profile Findings

On the author’s production profile, the tool reports a clean bill for all 34 checks. Notably, the same profile *does* carry plugins that use `ctx.inject(["settings"], cb)` (the safe runtime pattern), which the final P9 correctly recognizes — validating that the check distinguishes the safe pattern from the defective one [19].

6.5 Ecosystem Convergence

Three independent community tools — `dsh-plugin-doctor` [21], `dsh-diagnose` [22], and `dsh-doctor` — converged on the `dsh-doctor/v1` envelope: lowercase status, `{ok, checks:[{name,status,detail}]}` minimal subset, and a provenance `tool` field. A symptom→check mapping (16 symptom families → our 34 checks, with honest coverage marks including two documented gaps: sandbox denials and approval policy have no offline probe) is published in the `dsh-doctor` README [26], and a contract document pins the shape [27]. This convergence was driven by discussion threads [20], [32], [34] and is the concrete evidence that a shared contract can absorb independently built tools.

Table 2 — Independent diagnostic tools converging on the `dsh-doctor/v1` contract.

Table 2: Independent diagnostic tools converging on the `dsh-doctor/v1` contract.

Tool	Side	Checks	Envelope mode	Status vocabulary
dsh-doctor [26]	user (env/profile/session)	30 (25 built-in + 5 catalog)	<code>--json --envelope</code>	lowercase pass/warn/fail
dsh-plugin-doctor [21]	author (pre-publish) + profile tripwire	manifest/patch/entry/files/build/pack/install/profile-shadow	<code>v1.6.0 --profile --json</code>	lowercase pass/warn/fail
dsh-diagnose [22]	user (symptom, knowledge-anchored)	17 <code>dsh-symptom.*</code> families	<code>--doctor-json</code>	lowercase (adopted)

7. Discussion and Open Problems

7.1 Threats to Validity

Construct validity. The check inventory and failure taxonomy were induced from 18 community reports plus the tool’s own field history; both are subject to selection bias (reports that reached the discussion forum, not the full population of incidents). The “30 checks” count is a snapshot (v0.2.7) and grows as new reports arrive — we treat the inventory as data (Property 1), not as a fixed benchmark.

Internal validity. The fixture corpus is single-developer and synthetic: the good/bad fixtures encode the author’s interpretation of each failure class. Two mitigations: (i) the isolation property (target fails, others pass) is machine-checked in CI; (ii) several fixtures were cross-verified against independent implementations (boyin111-1’s sibling tool, dsh-plugin-doctor’s acceptance harness [32]). Cross-implementation fixture sharing remains partial.

External validity. dsh-doctor is evaluated on one harness (two consecutive release trains: 0.1.0-rc.6 and 0.1.5-rc.1) and one production profile. The check-lifecycle *model* (Section 5) is claimed as general, but its evidence base is a single ecosystem; generalization to other plugin-based harnesses (e.g., Koishi [5]) is untested. The environment checks are partially platform-dependent (verified on macOS; Windows-specific paths exercised only through reported cases [#1724], [#1965]).

Conclusion validity. The empirical claims are case-based (Section 6): real bugs caught and false positives eliminated are documented incidents, not randomized trials. We report precision/recall as open work rather than over-claiming them.

7.2 Reproducibility

- **Artifact:** the tool and all fixtures are in [26]; the paper’s claims map to the fixed commit `189f3e2` of `moonquake2004/dsh-doctor`.
- **Regression suite:** `node --test plugin/test/` (64 tests: 39 fixtures + 25 catalog) reproduces Section 6.1’s detection cases and the Section 6.2 false-positive eliminations.
- **Contract:** `docs/doctor-contract.md` [27] pins the envelope; `docs/check-lifecycle.md` [28] pins the model; `docs/paper-offline-diagnostics.md` is this paper.
- **Environment:** Node 18 (tested on 22), pnpm 9.15.9, macOS/Windows reports from the cited discussions.

7.3 Open Problems

Limitations. The checks are heuristic and name/regex-based; they detect *symptoms* (e.g., an inject list naming a client service) not *proofs*. Semantic boundaries reduce but cannot eliminate false positives; the certification gate (Property 3) is the only honest mitigation. Two documented coverage gaps — sandbox denials and approval policy — are runtime-only phenomena with no offline static probe, and we deliberately do not claim coverage there. The corpus is single-developer; cross-implementation fixture sharing (with boyin111-1’s sibling tool, for example) is ongoing.

The AI-generated plugin problem. A recurring theme in the reports is that harness agents themselves generate broken plugins (missing injects, client services as server dependencies, duplicate inserts) [17], [19]. This shifts the burden from “author discipline” to “automated authoring must be checked” — making offline diagnostics a *hard dependency* of the self-evolving-agent vision, and giving the check-lifecycle model an immediate consumer.

Open problems.

1. **Check registry governance.** Who hosts the shared check registry, and how do canonical ids coexist with tool-local ids (currently handled by the `tool` provenance field)? The registry-contract RFC [32] is the natural home, but checks-as-data (Property 1) needs a registry of its own.
2. **Certification matrix.** On which release trains must a fixture pair pass — latest only, or N-1 as well? This interacts directly with Property 2’s introspection: an anchor that passes on the current train is the strongest form of “still certified”.
3. **Signability.** Is pinned-repository-plus-read-only-probes sufficient trust for a remote catalog, or do rules need signing?
4. **Formal prevention.** The Cordis paper’s revertible effects and reactive coefficients [2] could, if fully realized, prevent whole failure classes (idempotent `register()`, host-first resolution) — the diagnostic layer exists today precisely because those guarantees are not yet there. We see Property 2 (introspection) as the bridge: a check that asserts a *contract* rather than a *value* is a runtime-verifiable claim about the harness’s own composability guarantees.

8. Conclusion

We presented the design and field experience of dsh-doctor, an offline diagnostic for a plugin-based agent harness, and generalized from it a check-lifecycle model. The model’s four properties — checks-as-data, introspection, fixture-based certification, and an evolution loop — are not aspirational; each is backed by a running implementation that has shipped 34 checks, maintained 109 regression tests, survived a three-round false-positive elimination campaign, and helped converge three independent community tools on one machine-readable contract. The broader lesson is that in a harness whose entire surface is composed at runtime, *diagnosis is a first-class engineering problem with its own lifecycle*, and that the formal composability guarantees the framework aspires to are the same properties a durable diagnostic layer must verify at runtime.

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